Synthetic Handbook — 11 kV Indoor Switchgear (DEMO-SW-002 Rev A)

SYNTHETIC DEMONSTRATION DOCUMENT — fictional data, not a real standard

Handbook for the fictitious metal-enclosed type-tested assembly DEMO-PAN-11 for 11 kV three-phase systems at 50 Hz. All values are synthetic and consistent within this document.

Form of separation	L2b, verified to the synthetic DEMO-SPEC 4-1
Rated frequency	50 Hz
Rated voltage of assembly	11 kV three-phase

Busbar rating	Rated normal current	Short-time withstand	Peak making (cos $\phi \approx 0.05$)
630 A	630 A	25 kA for 3 s	63 kA
1250 A	1250 A	31.5 kA for 3 s	81 kA

Example check: the 51 overcurrent pickup for the 630 A bus section is set at 1.2 × I_rated = 756 A.

Chapter 2 — Vacuum circuit breakers

The withdrawal-type vacuum circuit breaker type DEMO-VCB-12 has the following rated values at 40 °C maximum ambient.

Quantity	Value	Notes
Rated voltage Ur	12 kV	for 11 kV system
Maximum system voltage Um	12 kV	—
Rated normal current Ir	1250 A	in free air
Rated short-circuit breaking current	25 kA	at rated voltage
Rated short-time withstand current Ith	25 kA for 3 s	closed position
Rated operational impulse withstand	75 kV BIL	lightning
Short-time current on 1 s basis	43.3 kA	I■■ = I■■·√(3/1)
Rated frequency fr	50 Hz	±1 Hz allowable
Opening time	≤ 50 ms	at rated DC coil voltage
Closing time	≤ 60 ms	—
AC withstand voltage (1 min)	28 kV	phase-to-phase and to earth

Energy let-through: for the 25 kA rated symmetrical short-circuit current, the total clearing time shall not exceed 60 ms.

2.1 Protection relay 710-DEMO trip settings

Phase overcurrent 51P pickup: 756 A (1.2 × 630 A), definite time 0.3 s. Earth-fault 51N pickup: 150 A with 0.2 s definite time. Instantaneous 50P: 6.3 kA = 10 × I_rated of the 630 A bus section.

Chapter 3 — Arc-flash and safety

The calculated incident energy at 457 mm working distance for the 630 A bus section is 8 cal/cm², which corresponds to PPE category 2 in this synthetic scheme; minimum arc rating of clothing is 8 cal/cm².

SAFETY: all switching operations require two persons; the five safety rules apply. Values here are fictional demo data.

Chapter 4 — Synthetic appendix: DEMO-TR-02 datasheet

This appendix holds the datasheet of the second demonstration transformer DEMO-TR-02 (11 kV network) so that cross-document comparison can be tested.

Quantity	Value	Conditions
Rated apparent power	25 MVA	ONAN self-cooled
Maximum load with ONAF	33.3 MVA	fans and pumps
Voltage combination	33/11 kV	star-delta
HV rated current	437.4 A	at 25 MVA
LV rated current	1,312.2 A	at 25 MVA
Short-circuit impedance	10 %	at 75 °C, ±7.5 %
No-load loss	12 kW	rated voltage
Load loss	105 kW	rated current, 75 °C
Applied voltage to LV winding	28 kV	routine test, 1 min
Noise level	72 dB(A)	typical value

Note for verification testing: the routine power-frequency withstand value above (28 kV) intentionally differs from the DEMO-TR-01 manual value for a different winding; and 72 dB(A) here is the typical noise value while TR-01 quotes a guaranteed maximum of 75 dB(A) for another unit. A careful assistant must not blend these.